A CASE-STUDY BASED REPORT

ON

Product Demand Forecasting Using Ridge and Lasso Regression

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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Abstract

Product demand forecasting helps businesses predict future demand from historical sales data, supporting inventory planning, promotion targeting, and pricing decisions. This case study develops a demand forecasting system using Ridge Regression and Lasso Regression — regularized linear models — on a retail dataset of 504 records and 13 columns (500 rows after deduplication). Predictors include previous sales, product price, discount percent, advertising spend, seasonality, holidays, competitor pricing, stock availability, and promotions, with Demand_Units as the target.

The data pipeline cleans the raw Excel file (duplicate removal, missing-value imputation), encodes categoricals, and standardizes numerics inside a scikit-learn Pipeline to prevent leakage. An 80/20 train/test split (random_state=42) is used. Ridge (α=1.0) and Lasso (α=0.1) are trained and compared on MAE, MSE, RMSE, and R^2. Lasso achieved R^2 0.9087 (MAE 26.14) and Ridge R^2 0.9085 (MAE 26.18) on the test set. Previous_Sales_Units, Advertising_Spend, and Discount_Percent emerged as the strongest demand drivers. The trained models are served through a FastAPI prediction backend inside the DemandIQ analytics dashboard.

Introduction.....

Accurate demand forecasting is central to retail operations. Overstocking ties up working capital and risks obsolescence, while understocking causes lost sales and emergency replenishment costs. Traditional forecasting often relies on moving averages or planner judgment, which struggle when demand depends on many interacting drivers such as price, discounts, advertising, seasonality, holidays, and competitor prices.

This case study applies machine learning to the problem, comparing two regularized linear regressors — Ridge (L2 penalty) and Lasso (L1 penalty) — on the DemandIQ retail dataset. Regularized models are attractive here: they handle multicollinearity among overlapping signals (e.g., Season, Month, Holiday), resist overfitting on a modest-sized dataset, and remain interpretable so planners can see which levers move demand.

Background.....

Classical forecasting methods (moving averages, exponential smoothing, ARIMA) model demand as a pure time series and ignore rich cross-sectional drivers like promotions and competitor prices. Ordinary least-squares linear regression uses all features but is sensitive to multicollinearity, producing unstable coefficients. Complex ensembles (random forests, gradient boosting, LSTMs) capture nonlinearities but need more data and offer limited interpretability.

Penalized linear models sit between these extremes. Ridge regression adds an L2 penalty ($\alpha \cdot ||w||^2$) that shrinks coefficients and stabilizes them under collinearity. Lasso adds an L1 penalty ($\alpha \cdot ||w||_1$) that additionally performs automatic feature selection by driving weak coefficients to exactly zero. Both are controlled by a single hyperparameter α, making them practical and explainable for business forecasting.

Motivation for the Project

Retailers lose money on both sides of a bad forecast: excess stock holding costs on one side, stockouts and lost customers on the other. A data-driven forecast that quantifies the effect of price, discount, and advertising lets category managers plan inventory and promotions with evidence instead of intuition.

This project was motivated by the need for a forecasting method that is accurate enough to trust, simple enough to explain to non-technical stakeholders, and cheap enough to run — which led to Ridge and Lasso regression, packaged as the interactive DemandIQ dashboard so the results are usable, not just numbers in a notebook.

Overview of the Work Carried Out

The work was carried out end to end. (1) Data: the 504-row demand_dataset.xlsx was inspected, deduplicated to 500 rows, and profiled. (2) Preprocessing: a scikit-learn ColumnTransformer applies median imputation plus StandardScaler to numeric columns and most-frequent imputation plus OneHotEncoder to categoricals, fitted only on training data. (3) Modeling: Ridge (α=1.0) and Lasso (α=0.1, max_iter=10000) pipelines were trained on an 80/20 split. (4) Evaluation: MAE, MSE, RMSE, and R^2 were computed on the held-out test set and embedded in the dashboard. (5) Deployment: the fitted pipelines were saved with joblib and served by a FastAPI backend (/predict), with a multi-page Chart.js frontend for exploration, analytics, and live prediction.

Scope and Responsible-Use Statement

Scope: this study covers a single 500-row retail dataset, two regularized linear models and their comparison, and the DemandIQ dashboard that presents the results. It does not cover deep-learning forecasters, multi-step time-series modeling, or deployment at production scale.

Responsible use: the models produce statistical estimates, not guarantees — forecasts should supplement, not replace, planner judgment, especially around unprecedented events (new product launches, supply shocks). The dataset's size and characteristics limit how far the conclusions generalize; retraining on the retailer's own data is expected before business use.

Problem Statement

This is a supervised regression problem. Given a feature vector x describing a product-period context — Year, Month, Product category, Price, Discount_Percent, Advertising_Spend, Previous_Sales_Units, Season, Holiday flag, Competitor_Price,

Stock_Available, and Promotion flag — predict the continuous target y = Demand_Units. Success is measured with regression metrics (MAE, MSE, RMSE, R^2) on held-out data, and model coefficients are used to interpret which drivers matter most.

Case-Study Scenario

A retail business holds one year of product-month sales records with pricing, promotion, and seasonality information. The category manager wants to know, before each planning cycle, how many units of each product will be demanded — so that purchase orders, warehouse space, and promotional budgets can be set correctly. The available data is the 504-row DemandIQ dataset (13 columns); the deliverable is a trained, evaluated forecasting model plus an interactive dashboard the manager can actually use.

Central Case-Study Question

Can regularized linear regression (Ridge vs. Lasso) accurately forecast product demand from retail drivers, and which penalty — L2 shrinkage or L1 shrinkage with automatic feature selection — generalizes better on held-out data?

Formal Problem Definition

Let $X \in R^{n \times p}$ be the preprocessed feature matrix (n=500 product-periods, p predictors after encoding) and $y \in R^n$ the demand vector. Learn $f: X \rightarrow y$ minimizing squared error with a penalty: Ridge solves $\min_w ||y - Xw||^2 + \alpha||w||^2$ (α=1.0); Lasso solves $\min_w ||y - Xw||^2 + \alpha||w||_1$ (α=0.1). Models are fit on 80% of the data and evaluated on the remaining 20% using MAE, MSE, RMSE, and R^2. The model with the better test metrics is selected, and its coefficients are interpreted as demand drivers.

Objectives that Guide the Study

1. Build a leakage-free preprocessing pipeline (imputation, encoding, scaling) for the retail dataset.
2. Train Ridge and Lasso regressors with documented hyperparameters.
3. Compare the models rigorously using MAE, MSE, RMSE, and R^2 on a held-out test set.
4. Identify the strongest demand drivers from model coefficients.
5. Package the trained models behind a FastAPI prediction service inside the DemandIQ dashboard for interactive use.

Assumptions and Limitations

Assumptions: historical relationships between drivers and demand persist into the forecast horizon; the 500-row sample represents the product mix the model will be applied to; missing values are missing at random (justifying median/mode imputation).

Limitations: the modest dataset size restricts model complexity; no explicit timeseries structure (trend, autocorrelation) is modeled; hyperparameter search was limited (fixed α values rather than full grid search with cross-validation); results are specific to this dataset's characteristics and product mix.

Dataset and Exploratory Data Analysis

The dataset and its exploratory analysis are documented in the subsections below.

About the Dataset

The DemandIQ dataset (data/demand_dataset.xlsx) contains 504 records and 13 columns. The target is Demand_Units (units demanded per product-period). The 12 predictors cover Year, Month, Product category, Price, Discount_Percent, Advertising_Spend, Previous_Sales_Units, Season, Holiday flag, Competitor_Price, Stock_Available, and Promotion flag. Data-quality inspection found 4 exact duplicate rows (removed, leaving 500) and missing values in several numeric input columns (handled by imputation inside the pipeline).

Purpose of EDA

EDA was performed to understand the data before any modeling: to check distributions and ranges of each feature, to find missing values and duplicates, to spot invalid entries, and to measure which predictors correlate most strongly with Demand_Units — so that modeling choices (e.g., regularization to handle collinear Season/Month/Holiday signals) are grounded in the data rather than guessed.

Preparation Steps Actually Performed

1. Loaded demand_dataset.xlsx with pandas and inspected shape, dtypes, and descriptive statistics. 2. Removed 4 exact duplicate rows (504 → 500). 3. Built a ColumnTransformer: numeric columns → SimpleImputer(strategy='median') + StandardScaler; categorical columns → SimpleImputer(strategy='most_frequent') + OneHotEncoder(handle_unknown='ignore'). 4. Split 80/20 with train_test_split (test_size=0.2, random_state=42); the preprocessor is fitted only on the training fold to prevent leakage. 5. Wrapped preprocessing and regressor in a single Pipeline for each model.

EDA Exploration - Implementation

Exploration was implemented in Python (pandas) alongside the training script backend/train_model.py. Missing-value counts per column were tabulated; numeric gaps were filled with column medians and categorical gaps with modes. Correlation analysis showed Previous_Sales_Units, Advertising_Spend, and Discount_Percent as the strongest positive correlates of Demand_Units, while Season, Month, and Holiday flags carried overlapping (collinear) seasonal signals — confirming the choice of regularized models. No extreme invalid outliers requiring removal were found beyond the imputed missing entries.

Summary of EDA Findings

Four duplicate rows were removed (504 → 500 records). Missing numeric values were median-imputed and categorical values mode-imputed inside the pipeline. Previous_Sales_Units, Advertising_Spend, and Discount_Percent emerged as the strongest demand correlates. Overlapping Season/Month/Holiday signals confirmed multicollinearity, justifying Ridge/Lasso over ordinary least squares. The cleaned 500row dataset was deemed suitable for regularized linear modeling.

Algorithm Selection and Implementation.....

Ridge and Lasso were selected because the EDA revealed multicollinearity (Season/Month/Holiday overlap), which ordinary least squares handles poorly, and because the business needs interpretable coefficients — ruling out black-box ensembles for this case study. Ridge (α=1.0) applies L2 shrinkage, keeping all features with stabilized coefficients. Lasso (α=0.1, max_iter=10000) applies L1 shrinkage, additionally zeroing out weak predictors for automatic feature selection.

Implementation (backend/train_model.py): a scikit-learn Pipeline chains the ColumnTransformer preprocessor with the regressor; both pipelines are fit on the 80% training split and evaluated on the 20% test split with MAE, MSE, RMSE, and R^2; fitted pipelines are persisted with joblib to backend/models/ridge.joblib and lasso.joblib. A FastAPI service (backend/app.py) loads the pipelines and exposes POST /predict, which the dashboard's Demand Prediction page calls — falling back to a clearly labeled demo estimate when the backend is offline.

Test results: Lasso — MAE 26.14, MSE 1032.75, RMSE 32.14, R^2 0.9087. Ridge — MAE 26.18, MSE 1034.35, RMSE 32.16, R^2 0.9085. Lasso marginally outperformed Ridge on all four metrics, so it is the selected model.

Conclusion

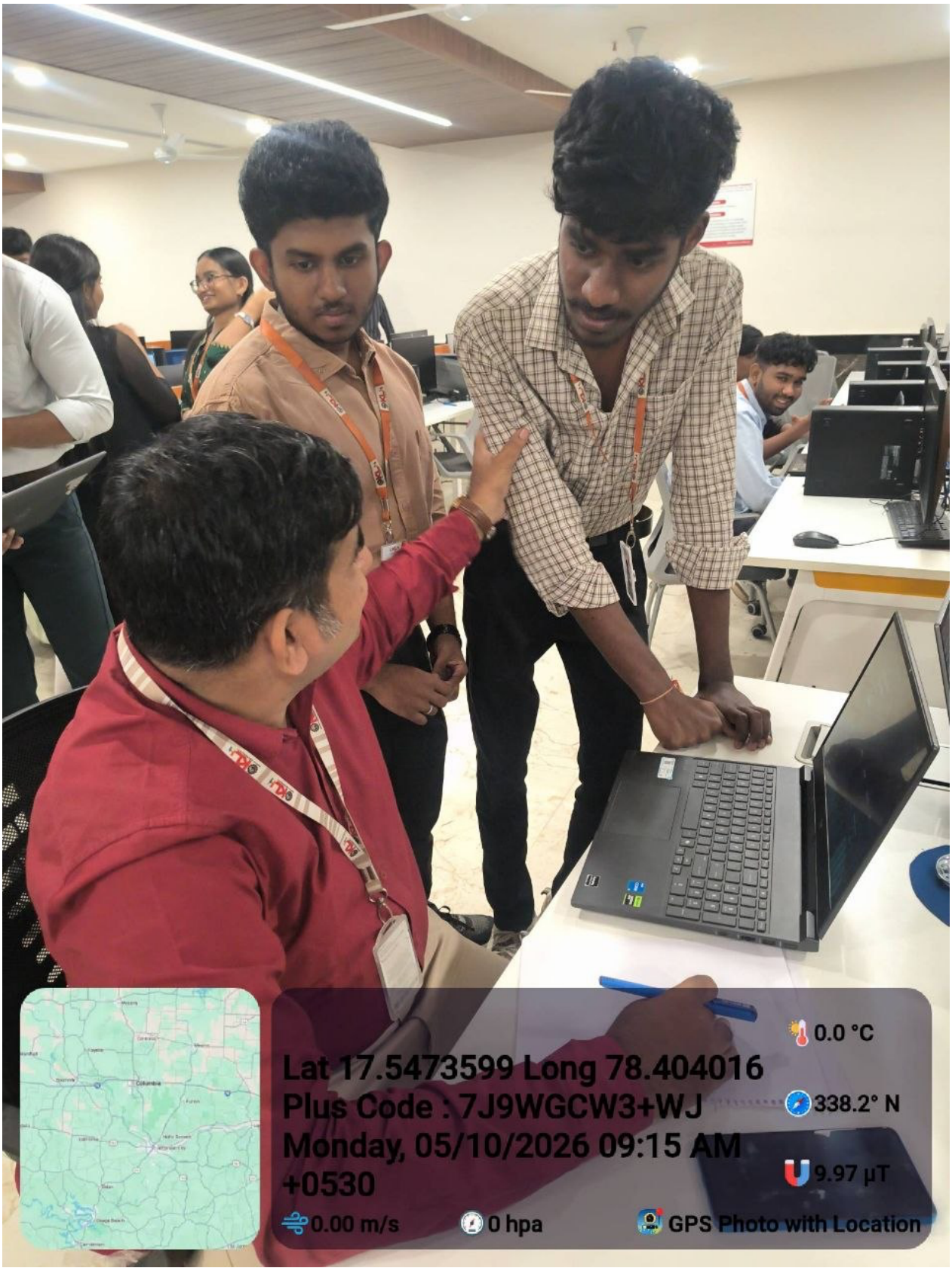
Both regularized models forecast product demand accurately on held-out data ($R^2 \approx 0.909$), with Lasso (α=0.1) marginally the best across MAE, MSE, RMSE, and R^2. Previous_Sales_Units, Advertising_Spend, and Discount_Percent were identified as the strongest demand drivers, giving planners concrete levers. The DemandIQ dashboard — with dataset exploration, analytics charts, model comparison, live prediction via FastAPI, and export — turns the models into a usable tool rather than a notebook experiment. Future work: larger hyperparameter search with crossvalidation, time-series features, and retraining on the retailer's own data before business use.

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